

Project #1 guidelines for Python

If you decide to write your project using python, I'll tell you what to do in project #1 including downloading necessary libraries and where to implement the code in project #1 framework.

Download necessary libraries

In project #1, to get the predict output of accuracy, you must install numpy, pandas, and scikit-learn. (Only use sklearn.metrics for performance measurement)

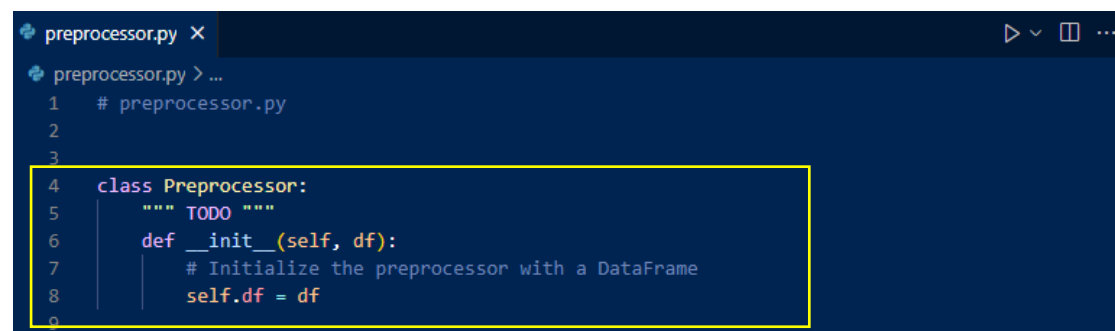
You can also use the *environment.yml* I provide for you to create conda environment.

Implementation

The detail of where you should implement the code is explaining here.

1. Complete ***Preprocessor class*** in *preprocessor.py*

Write your custom data preprocessing functions as the methods in Preprocessor class.



```
preprocessor.py X
preprocessor.py > ...
1  # preprocessor.py
2
3
4  class Preprocessor:
5      """ TODO """
6      def __init__(self, df):
7          # Initialize the preprocessor with a DataFrame
8          self.df = df
9
```

2. Complete **DataPreprocessing function** in *main.py*
Use the custom methods you implemented to complete dataPreprocessing function.

```
main.py X
main.py > ...
1  import numpy as np
2  import pandas as pd
3
4  from preprocessor import Preprocessor
5  from model import LogisticRegressionClassifier
6  from sklearn.metrics import accuracy_score
7
8  def dataPreprocessing():
9      """ TODO, implement your own dataPreprocess function here. """
10
11     train_X = None
12     train_y = None
13     test_X = None
14     test_y = None
15     return train_X, train_y, test_X, test_y
16
17
18 def main():
19     train_X, train_y, test_X, test_y = dataPreprocessing()
20     model = LogisticRegressionClassifier()
21     model.fit(train_X, train_y)
22
23     pred = model.predict(test_X)
24     prob = model.predict_proba(test_X)
25     prob = [f'{x:.5f}' for x in prob]
26     # print(f'Prob: {prob}')
27     print(f'Acc: {accuracy_score(pred, test_y):.5f}')
28
29
30 if __name__ == "__main__":
31     np.random.seed(0)
32     main()
```

Note: the input of *model.fit()* should be **numpy array**, you may need to use *to_numpy()* to convert your dataframe to numpy array.